

REFERME INVESTMENTS

Business Analysis Project Report & Documentation

1. Executive Summary

ReferMe Investments required a consistent, data-driven way to evaluate service businesses as potential investment targets. The solution was framed as an investment analytics system rather than an analysis of a single operating company. The project defines 28 KPIs across Financial Strength, Operational Performance, Growth, Customer Health and Risk, feeds those measures into dimension scores and an overall Investment Score, and translates the framework into Power BI decision-support views.

Business Need	BA Response
Inconsistent or subjective investment evaluation	Standardized five-dimension assessment framework
Many financial/operational measures without decision context	28-KPI catalogue tied to investment decisions
Need to compare different service businesses	Normalized 0-100 scoring architecture
Need to identify weak/risky targets quickly	Risk indicators and alert concepts
Need post-investment visibility	Portfolio monitoring with actual-vs-target and score/risk changes

1.1 Project Objective

Design an investment analytics framework with 25+ KPIs and a multi-dimensional scoring model covering financial strength, operational performance, growth, customer health and risk, supported by Power BI dashboards that translate underwriting methodology into a measurable evaluation system.

1.2 Success Criteria

Every KPI, chart and dataset must support a better, more consistent investment decision.

Investment targets must be comparable through common definitions and scoring logic.

Decision makers must be able to drill from portfolio view to individual business and underlying KPIs.

Risk deterioration must be visible through monitoring and early-warning indicators.

Methodology assumptions must remain transparent and subject to stakeholder validation.

2. Project Scope

In Scope	Out of Scope / Not Established by Source
Investment evaluation framework for service businesses	Actual proprietary ReferMe underwriting rules
28 KPI definitions and five scoring dimensions	Final approved production scoring thresholds
Illustrative 0-100 weighted scoring methodology	Claims that sample business data represents real ReferMe portfolio data
Power BI requirements for five decision pages	Production deployment architecture
Investment evaluation and portfolio monitoring concepts	Legal, tax, valuation or investment advice

3. Stakeholder Analysis

Stakeholder	Primary Interest	BA Engagement
Investment / Underwriting Team	Consistent investment decisions	Elicit decision criteria, validate KPIs and scoring
Senior Management	Portfolio quality and strategic visibility	Executive requirements, scorecards and recommendations
Finance / Investment Analysts	Financial health and cash generation	KPI definitions, calculation rules, benchmark needs
Operations / Portfolio Team	Operating efficiency and post-investment performance	Operational KPIs, targets and monitoring requirements
Data / BI Team	Reliable data model and dashboards	Source-to-KPI mapping, acceptance criteria, data quality rules
Investment Target / Service Business	Clear information requirements	Data request definitions and business-profile inputs

4. Business Requirements Document (BRD)

4.1 Business Requirements

ID	Requirement	Priority
BR-01	System shall evaluate each investment target across five dimensions: Financial Strength, Operational Performance, Growth, Customer Health and Risk.	Must
BR-02	System shall calculate and display at least 25 investment-relevant KPIs; target framework contains 28.	Must
BR-03	System shall convert KPI performance into comparable dimension scores.	Must
BR-04	System shall produce an overall 0-100 Investment Score using validated dimension weights.	Must
BR-05	System shall support an investment recommendation outcome such as Invest, Review or Reject once thresholds are approved.	Must
BR-06	Users shall be able to inspect underlying KPIs behind a business score.	Must
BR-07	System shall provide portfolio-level monitoring of actual vs target, KPI trends, score changes and risk changes.	Must

BR-08	System shall surface early-warning indicators for material deterioration or concentration risks.	Should
BR-09	Dashboard shall support business-level drill-down across revenue, customers, profitability, operations, growth and risk.	Must
BR-10	Data quality shall be measured as part of the risk framework.	Should

4.2 Business Rules / Assumptions

All scoring weights and thresholds require stakeholder validation before production use.

Higher risk must not be interpreted as a positive score without an explicit normalization rule.

Missing or low-quality data should reduce confidence in the recommendation and be visible to decision makers.

The sample service-business dataset is an analytical scenario for testing the framework, not asserted as ReferMe's actual portfolio.

Recommendation labels should only be finalized after business owners approve score thresholds and override rules

5. Functional Requirements Specification (FRS)

ID	Functional Requirement
FR-01	Maintain investment-target master data including business ID/type and evaluation attributes.
FR-02	Calculate KPI values from validated financial, operational, customer and risk inputs.
FR-03	Normalize KPI values to a common scoring scale according to approved thresholds.
FR-04	Calculate five dimension scores using approved KPI weights.
FR-05	Calculate overall Investment Score using approved dimension weights.
FR-06	Display investment pipeline KPIs including businesses evaluated, average score and decision counts.
FR-07	Provide individual Investment Evaluation view with overall and dimension scores.
FR-08	Provide Business Profile trends for revenue, customers, profitability, operations, growth and risk.
FR-09	Provide Portfolio Monitoring for actual vs target, trends, score changes and early warnings.
FR-10	Provide Risk & Alerts view for revenue decline, churn, margin deterioration, concentration and cash-flow performance.
FR-11	Allow filtering/drill-down by investment target and relevant business dimensions.
FR-12	Expose data-quality status to prevent false confidence in incomplete evaluations.

6. KPI Catalogue

Dimension	KPI	Business Meaning / Calculation Intent
Financial Strength	Revenue	Scale of business / top-line performance
Financial Strength	Revenue Growth %	Change in revenue over comparison period
Financial Strength	Gross Margin %	Gross profit as % of revenue
Financial Strength	EBITDA Margin %	EBITDA as % of revenue
Financial Strength	Operating Cash Flow	Cash generated from operations
Financial Strength	Cash Flow Margin %	Operating cash flow as % of revenue
Operational Performance	Revenue per Employee	Revenue / employee count
Operational Performance	Operating Cost %	Operating costs as % of revenue
Operational Performance	Labor Cost %	Labor cost as % of revenue
Operational Performance	Capacity Utilization %	Used capacity / available capacity
Operational Performance	On-Time Service %	On-time services / total services
Operational Performance	Operational Efficiency Score	Composite operating-efficiency measure
Growth	Customer Growth %	Change in active customer base
Growth	New Customer Acquisition	New customers acquired in period
Growth	Recurring Revenue %	Recurring revenue / total revenue
Growth	Average Revenue per Customer	Revenue / active customers

Growth	12-Month Revenue CAGR	Annualized revenue growth
Customer Health	Customer Retention %	Retained customers / eligible prior customers
Customer Health	Customer Churn %	Lost customers / eligible prior customers
Customer Health	Repeat Customer Rate	Repeat customers / total customers
Customer Health	Customer Complaint Rate	Complaints / customer or service base
Customer Health	Customer Concentration %	Revenue exposure to largest customer(s)
Risk	Revenue Volatility	Variability of revenue over time
Risk	Customer Concentration Risk	Risk derived from customer dependency
Risk	Margin Deterioration	Negative movement in margin
Risk	Cash Flow Risk	Risk arising from weak/unstable operating cash flow
Risk	Data Quality Score	Completeness/validity/consistency of required data
Risk	Overall Business Risk Score	Composite business-risk measure

7. Scoring Model

The source framework proposes a 0-100 scoring system. As an example, Financial Strength may combine Revenue Growth (25%), EBITDA Margin (20%), Cash Flow Margin (20%), Gross Margin (15%), Recurring Revenue (10%) and Revenue Stability (10%). The final Investment Score then combines five dimension scores.

Dimension	Proposed Weight
Financial Strength	25%
Operational Performance	20%
Growth	20%
Customer Health	15%
Risk	20%
Total	100%

Proposed formula: Investment Score = Σ (Dimension Score $\times$ Approved Dimension Weight).

Control: these weights are methodology assumptions and must not be represented as ReferMe's proprietary approved rules until validated.

8. User Stories & Acceptance Criteria

ID	User Story	Acceptance Criteria
US-01	As an investment analyst, I want to view all businesses with their investment scores so that I can prioritize due diligence.	Given evaluated businesses exist, when I open the overview, then I can see businesses evaluated, average score and decision counts.
US-02	As an underwriter, I want to drill into a target's five dimension scores so that I	When a business is selected, then the overall score, five dimension scores and underlying KPIs are visible.

	can understand the drivers of the recommendation.	
US-03	As a portfolio manager, I want actual-vs-target and score trends so that I can detect post-investment deterioration.	For an invested business, the monitoring view shows actual vs target, KPI trend, score change and risk change.
US-04	As a risk reviewer, I want alerts for declining revenue, high churn, margin deterioration and concentration so that I can investigate early.	When an approved threshold is breached, the affected business and indicator are highlighted for review.
US-05	As a decision maker, I want data-quality visibility so that I can distinguish weak performance from weak evidence.	When required inputs are incomplete or invalid, the Data Quality Score/status is visible alongside the evaluation.

9. Process Models

9.1 AS-IS (Conceptual)

Investment target submitted → Financial/operational/customer information collected →
 Analysts review measures → Findings interpreted → Investment decision prepared.

9.2 TO-BE

Investment target submitted → Standardized data intake → Data-quality validation → 28 KPI calculation → KPI normalization → Five dimension scores → Overall Investment Score → Invest/Review/Reject decision support → Portfolio monitoring → Risk alerts.

9.3 Gap Analysis

Gap	Current Risk	Target-State Change
Non-standard evaluation criteria	Different analysts may emphasize different metrics	Common five-dimension framework
Metrics not linked to decisions	Dashboard can become descriptive only	KPI-to-score-to-recommendation traceability
Limited comparability	Businesses of different profiles difficult to compare	Normalized 0-100 scoring
Weak post-investment visibility	Deterioration detected late	Portfolio trends and early-warning indicators
Data completeness uncertainty	False confidence in scores	Data Quality Score and validation controls

10. Data Requirements & Data Dictionary

Data Domain	Representative Fields	Used For
Business Master	Business ID, type, evaluation status	Target identification and filtering
Financial	Revenue, gross profit, EBITDA, operating cash flow, costs	Financial Strength / Risk
Workforce & Operations	Employees, labor cost, capacity, on-time service	Operational Performance
Customers	Customer count, new/lost/repeat customers, complaints, customer revenue	Growth / Customer Health / concentration

Time	Month/quarter/year	Trends, CAGR, volatility, actual vs target
Scoring	KPI score, dimension score, investment score, recommendation	Decision support
Data Quality	Missing, invalid, duplicate or stale input indicators	Data Quality Score / confidence

11. Power BI Requirements

Page	Decision Question	Required Content
1. Investment Overview	What does the investment pipeline look like?	Businesses Evaluated; Average Investment Score; Invest/Review/Reject; Average Risk, Growth and Financial Strength
2. Investment Evaluation	Should ReferMe invest in this business?	Overall score; five dimension scores; underlying KPIs
3. Business Profile	What is happening inside this target?	Revenue trend, customers, profitability, operations, growth and risk
4. Portfolio Monitoring	How are investments performing after investment?	Actual vs target; KPI trends; score/risk changes; early warnings; attention list
5. Risk & Alerts	Where should the investment team investigate?	Revenue decline, churn, margin deterioration, concentration and strong cash-flow signals

12. Requirements Traceability Matrix (RTM)

BR	Functional Req.	User Story	UAT	Output
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BR-01	FR-04	US-02	UT-02	Power BI Page 2
BR-02	FR-02	US-02	UT-01	Pages 2-3
BR-03	FR-03/04	US-02	UT-02	Page 2
BR-04	FR-05	US-01/02	UT-03	Pages 1-2
BR-05	FR-05	US-01	UT-04	Page 1
BR-06	FR-07	US-02	UT-05	Page 2
BR-07	FR-09	US-03	UT-06	Page 4
BR-08	FR-10	US-04	UT-07	Page 5
BR-09	FR-08	US-02	UT-08	Page 3
BR-10	FR-12	US-05	UT-09	Pages 2/5

13. UAT Plan & Test Cases

UAT objective: validate that the analytical framework produces understandable, traceable and decision-useful outputs using approved sample/test data. Numeric thresholds and recommendation cut-offs require business approval before final UAT.

Test ID	Scenario	Test Data / Action	Expected Result
UT-01	KPI calculation	Load complete valid test data	All applicable KPIs calculate with no unexpected blanks
UT-02	Dimension scoring	Provide KPI scores and approved weights	Dimension score equals weighted KPI result
UT-03	Investment scoring	Provide five dimension scores	Overall score equals approved weighted calculation
UT-04	Recommendation	Use test scores around approved thresholds	Correct Invest/Review/Reject label returned

UT-05	Drill-down	Select one investment target	Underlying dimension/KPI values reconcile to summary
UT-06	Portfolio trend	Load multiple periods	Actual-vs-target and score/risk changes display correctly
UT-07	Risk alert	Create threshold-breach scenario	Relevant warning appears for affected business
UT-08	Business profile	Select target	Financial, customer, operations, growth and risk trends filter correctly
UT-09	Data quality	Remove/invalid required inputs	Data-quality status declines and limitation is visible

14. Risk, Issue, Assumption & Dependency Log (RAID)

Type	Item	Impact	Mitigation / Action
Risk	Scoring weights not validated	Misleading investment ranking	Run stakeholder calibration and sensitivity testing
Risk	Different business models have different benchmarks	Unfair cross-business comparison	Use segment-aware benchmarks where required
Risk	Incomplete source data	Weak KPI reliability	Data-quality controls and confidence flag
Risk	Composite score hides critical weakness	False positive investment decision	Always expose dimension/KPI

			drill-down and override rules
Assumption	Required financial/operational/customer data can be obtained	Framework depends on adequate inputs	Validate data availability during discovery
Assumption	0-100 scoring is acceptable for decision support	Enables common comparison	Confirm with investment stakeholders
Dependency	Power BI data model and refresh	Dashboard usability	Define source mapping and refresh ownership
Dependency	Business-approved thresholds	Recommendations and alerts	Do not productionize labels until approved

15. Change Control

Any requested change to KPI definitions, weights, thresholds, recommendation logic or dashboard decision views should be recorded as a change request. The BA should document business rationale, affected requirements/KPIs, data impact, scoring impact, testing impact, owner, approval and target release before implementation.

Change Request Fields
CR ID; requestor; date; requested change; rationale; impacted BR/FR/KPI; scoring impact; data impact; dashboard impact; test impact; priority; approver; status

16. Project Plan: February - June 2026

The source document defines the target framework but does not provide dated delivery events. The following phase plan is therefore a reconstructed BA project plan for the requested Feb-Jun 2026 reporting period, not a claim of actual dated completion.

Period	BA Phase	Key Activities	Primary Artefacts
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February 2026	Discovery & Scope	Clarify investment decision problem; identify stakeholders; define five dimensions; establish scope and assumptions	Project charter, stakeholder map, scope, AS-IS
March 2026	Requirements & KPI Design	Elicit decision criteria; define 28 KPIs; document calculations/data needs; prioritize requirements	BRD, KPI catalogue, data dictionary, TO-BE, gap analysis
April 2026	Scoring & Functional Design	Design 0-100 methodology; define dimension weighting concept; translate needs into functional requirements/user stories	FRS, scoring specification, user stories, acceptance criteria, RTM
May 2026	BI Design & Validation	Define five Power BI pages; map KPI-to-visual requirements; prepare test scenarios; review data-quality and risk logic	Dashboard requirements, wireframe specification, UAT plan, RAID log
June 2026	UAT, Handover & Closure	Validate calculations and decision flows using test data; log issues/changes; prepare handover and project narrative	UAT cases/results template, change log, handover, project report

17. Deliverables Register

Deliverable	Purpose	Status
Project Charter / Executive Summary	Business objective and value	Included
Stakeholder Analysis	Stakeholder needs and engagement	Included
BRD	Business requirements	Included
FRS	Functional requirements	Included
KPI Catalogue	28 KPI definitions/intents	Included
Scoring Model Specification	0-100 methodology and weights	Included as proposed methodology
User Stories & Acceptance Criteria	Agile delivery requirements	Included
AS-IS / TO-BE / Gap Analysis	Process transformation	Included
Data Requirements / Dictionary	Data needed for KPIs	Included
Power BI Requirements	Decision-layer design	Included
RTM	End-to-end traceability	Included
UAT Plan & Test Cases	Business validation	Included
RAID Log	Delivery governance	Included
Change Control	Controlled requirement changes	Included
Project Timeline	Feb-Jun delivery plan	Included as reconstructed plan
Closure / Handover	Completion and ownership checklist	Included

18. Definition of Done

All approved BRD and functional requirements are traceable to test cases and dashboard outputs.

All 28 KPI definitions, sources, formulas and ownership are validated.

Scoring weights, thresholds and recommendation logic receive explicit business approval.

Dashboard figures reconcile to source/test data.

Critical UAT defects are closed or formally accepted.

Data-quality limitations and methodology assumptions are documented.

Business owners receive the final documentation, dashboard requirements and operating guidance.

19. Project Closure & Handover

Closure Item	Evidence / Handover
Requirements baseline	Approved BRD/FRS and change log
Analytical methodology	KPI catalogue + scoring specification
Decision layer	Power BI page requirements and filters
Quality assurance	RTM + UAT evidence
Operational ownership	Named owners for data refresh, KPI definitions and threshold changes
Known limitations	Assumptions, data gaps and unresolved risks
Future enhancements	Benchmark calibration, scenario/sensitivity analysis, automated alerting, model governance

20. Project Summary

I worked on an investment analytics project for ReferMe Investments focused on making service-business evaluation more consistent and measurable. I translated the investment

decision process into five analytical dimensions and a 28-KPI framework covering financial strength, operations, growth, customer health and risk. From a BA perspective, I structured business and functional requirements, KPI definitions, user stories, acceptance criteria, process flows, traceability and UAT scenarios. I also defined the requirements for a multi-dimensional 0-100 scoring model and Power BI views for investment evaluation, business profiling, portfolio monitoring and risk alerts. A key control was keeping proposed scoring assumptions transparent so that weights and thresholds could be validated by business stakeholders before production use.

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